WASP-10b

R-1489

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-10_131125` · created 2026-08-01T13:40:55+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456621.7387 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.21 +/- 0.02
Transit depth (Rp/Rs)^2	4.42 +/- 0.84 [%]
Orbital Inclination (inc)	86.63 +/- 1.13
Airmass coefficient 1 (a1)	1.073 +/- 0.039
Airmass coefficient 2 (a2)	-0.04 +/- 0.014
Scatter in the residuals of the lightcurve fit is	3.71 %
Best Comparison Star	#3 - [295, 260]
Optimal Aperture	2.39
Optimal Annulus	11.37
Transit Duration (day)	0.08 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.21 ± 0.02 vs 0.1592 (deviation 2.13σ)
Duration fit vs published	0.08 d vs 0.0946 d (deviation 0.81σ)
Mid-time O–C	5.2 min
Depth SNR	8.8
Residual scatter	3.71 %

Light curve

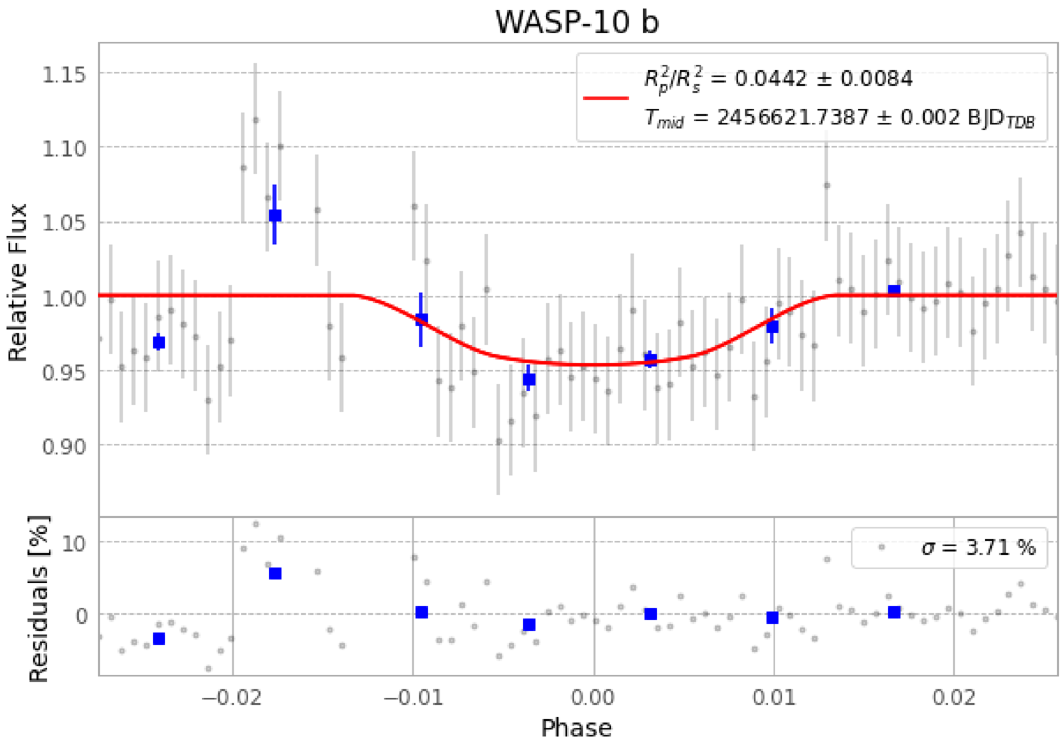

Figure 1 — FinalLightCurve_WASP-10 b_2013-11-24.png (SHA-256 a9605c482cb3317c…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [353, 304], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 0f69a42432320078…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

80 FITS frames (53 MB), `WASP-10131125033612.FITS` → `WASP-10131125073312.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-131125.log (5d8d7752166e…), FinalLightCurve_WASP-10 b_2013-11-24.png (a9605c482cb3…), FinalLightCurve_WASP-10 b_2013-11-24.csv (3507f63f6a05…)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)